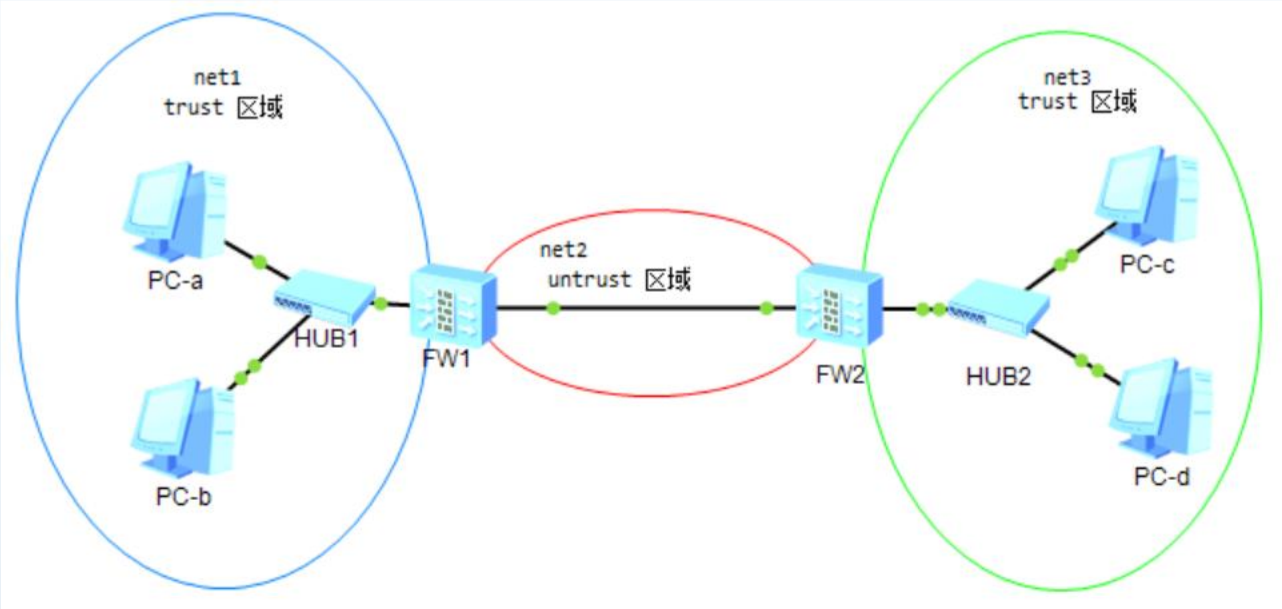
实验2 防火墙基础配置

本章目标

- 通过两个防火墙，互联两个网络，实现互通
- 掌握接口ip地址配置
- 掌握静态路由配置

1 实验规划

设备与拓扑



实验设备：
USG6000：2台
HUB：2台
PC： 4台

数据规划

网络设备名称	接口	IP地址	所属安全区域	互联对端设备	对端接口
FW1	G1/0/0	2.x.y.1/24	untrust	FW2	G1/0/0
	G1/0/1	1.x.y.254/24	trust	HUB1	e0/0/0
FW2	G1/0/0	2.x.y.2/24	untrust	FW2	G1/0/0
	G1/0/1	3.x.y.254/24	trust	HUB2	e0/0/0
HUB1	e0/0/0	无	无	FW1	G1/0/1
	e0/0/1	无	无	PC-a	
	e0/0/2	无	无	PC-b	
HUB2	e0/0/0	无	无	FW2	
	e0/0/1	无	无	PC-c	
	e0/0/2	无	无	PC-d	

主机名称	IP地址	子网掩码	默认网关
PC-a	1.x.y.1	255.255.255.0	1.x.y.254
PC-b	1.x.y.2	255.255.255.0	1.x.y.254
PC-c	3.x.y.1	255.255.255.0	3.x.y.254
PC-d	3.x.y.2	255.255.255.0	3.x.y.254

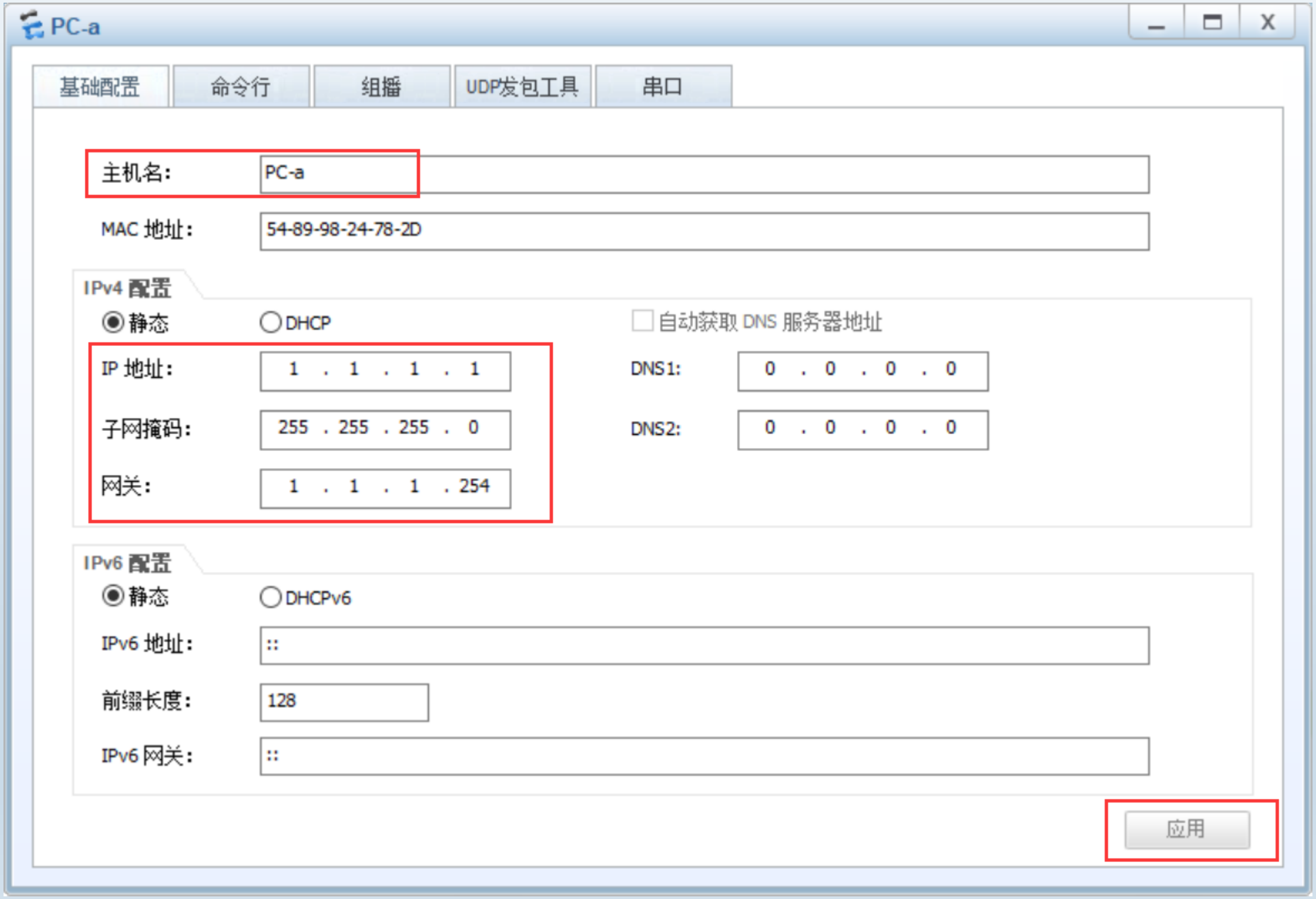
特别注意：IP地址中的x,y的含义
x：学号倒数第三位； y:学号倒数后两位

2 实验步骤

实验步骤

- 构建拓扑
- 启动设备
- 配置PC的IP地址和默认网关
- 配置防火墙
- 结果验证

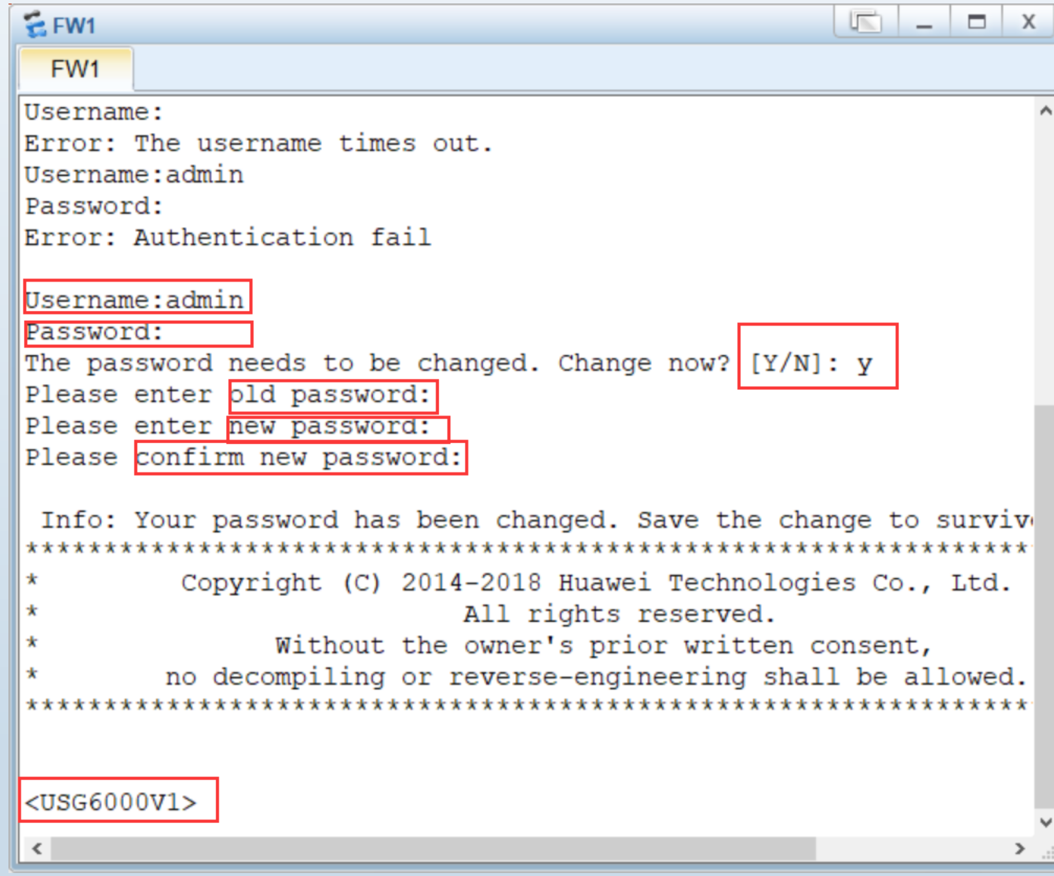
配置PC



PC-a
基础配置
命令行
组播
UDP发包工具
串口
主机名:
PC-a
MAC 地址:
54-89-98-24-78-2D
IPv4 配置
静态
DHCP
自动获取 DNS 服务器地址
IP 地址:
1 . 1 . 1 . 1
DNS1:
0 . 0 . 0 . 0
子网掩码:
255 . 255 . 255 . 0
DNS2:
0 . 0 . 0 . 0
网关:
1 . 1 . 1 . 254
IPv6 配置
静态
DHCPv6
IPv6 地址:
::
前缀长度:
128
IPv6 网关:
::
应用

启动设备-首次登录

- 防火墙首次启动登录需要验证用户身份
- 采用系统初始默认用户名和密码
- 首次登录后需要立即修改密码
- 修改为统一密码，方便日后重复调试



设备首次开启

- 默认用户名：admin
- 默认密码：Admin@123
- 设置新密码：admin@123

华为防火墙命令行模式

- sysname：配置设备名称
- Interface g0/0/0：进入接口配置模式
- display ：用于在任何视图下显示相关内容
- Ctrl+z 快捷方式，退出到<> 用户模式
- save: 在用户视图模式下完成配置文件的保存
- display ip routing-table：显示路由表
- display zone：显示区域
- display firewall session table：显示防火墙会话表
- display security-policy all（自定义名称）：显示安全策略

配置防火墙FW1

- **修改设备名称**
  - sysname 学号后三位-1
- **配置接口地址**
  - interface GigabitEthernet1/0/0（配置接口G1/0/0)
  - ip address 2.x.y.1 255.255.255.0(配置IP地址)
  - service-manage ping permit（开启ping 服务）
  - interface GigabitEthernet1/0/1
  - ip address 1.x.y.254 255.255.255.0
  - service-manage ping permit

配置防火墙FW1

- **配置安全区域**
  - firewall zone trust
  - add interface GigabitEthernet1/0/1
  - firewall zone untrust
  - add interface GigabitEthernet1/0/0
- **配置缺省安全策略**
  - security-policy
  - default action permit
- **配置缺省缺省路由**
  - ip route-static 0.0.0.0 0.0.0.0 2.x.y.2
- **保存配置**
  - 在普通用户模式下，save

3 结果验证

结果验证

❖ 查看接口信息

```
[test1-a]disp ip interface brief
2023-04-03 14:49:58.560
*down: administratively down
^down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
(E): E-Trunk down
The number of interface that is UP in Physical is 5
The number of interface that is DOWN in Physical is 5
The number of interface that is UP in Protocol is 4
The number of interface that is DOWN in Protocol is 6

Interface                         IP Address/Mask      Physical   Protocol
GigabitEthernet0/0/0              192.168.0.1/24       down       down
GigabitEthernet1/0/0              2.2.2.1/24           up         up
GigabitEthernet1/0/1              1.1.1.254/24         up         up
GigabitEthernet1/0/2              unassigned           up         down
GigabitEthernet1/0/3              unassigned           down       down
GigabitEthernet1/0/4              unassigned           down       down
GigabitEthernet1/0/5              unassigned           down       down
GigabitEthernet1/0/6              unassigned           down       down
NULL0                             unassigned           up         up(s)
Virtual-if0                       unassigned           up         up(s)
```

结果验证

❖ 查看安全区域

```
[test1-a]display zone interface
2023-04-03 14:52:54.900
local
 interface of the zone is (0):
#
trust
 interface of the zone is (2):
    GigabitEthernet0/0/0
    GigabitEthernet1/0/1
#
untrust
 interface of the zone is (1):
    GigabitEthernet1/0/0
#
dmz
 interface of the zone is (0):
#
```

结果验证

❖ 查看路由表

```
[test1-a]disp ip routing-table
2023-04-03 14:55:03.960
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 7        Routes : 7

Destination/Mask    Proto   Pre  Cost        Flags NextHop         Interface

        0.0.0.0/0   Static  60   0             RD   2.2.2.2         GigabitEthernet
1/0/0
        1.1.1.0/24  Direct  0    0              D   1.1.1.254       GigabitEthernet
1/0/1
      1.1.1.254/32  Direct  0    0              D   127.0.0.1       GigabitEthernet
1/0/1
        2.2.2.0/24  Direct  0    0              D   2.2.2.1         GigabitEthernet
1/0/0
        2.2.2.1/32  Direct  0    0              D   127.0.0.1       GigabitEthernet
1/0/0
      127.0.0.0/8   Direct  0    0              D   127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct  0    0              D   127.0.0.1       InLoopBack0
```

结果验证

❖ 查看路由表

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1/0/0
        1.1.1.0/24  Direct  0    0              D   1.1.1.254       GigabitEthernet
1/0/1
      1.1.1.254/32  Direct  0    0              D   127.0.0.1       GigabitEthernet
1/0/1
        2.2.2.0/24  Direct  0    0              D   2.2.2.1         GigabitEthernet
1/0/0
        2.2.2.1/32  Direct  0    0              D   127.0.0.1       GigabitEthernet
1/0/0
      127.0.0.0/8   Direct  0    0              D   127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct  0    0              D   127.0.0.1       InLoopBack0
```

验证网络连通性

❖ 在PC-a上分别ping网关、PC-c和PC-d的IP地址

```
PC>ping 1.1.1.254

Ping 1.1.1.254: 32 data bytes, Press Ctrl_C to break
From 1.1.1.254: bytes=32 seq=1 ttl=255 time<1 ms
From 1.1.1.254: bytes=32 seq=2 ttl=255 time=16 ms
From 1.1.1.254: bytes=32 seq=3 ttl=255 time<1 ms
From 1.1.1.254: bytes=32 seq=4 ttl=255 time<1 ms
From 1.1.1.254: bytes=32 seq=5 ttl=255 time=15 ms

--- 1.1.1.254 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 0/6/16 ms
```

```
PC>ping 3.3.3.1

Ping 3.3.3.1: 32 data bytes, Press Ctrl_C to break
Request timeout!
From 3.3.3.1: bytes=32 seq=2 ttl=126 time=16 ms
From 3.3.3.1: bytes=32 seq=3 ttl=126 time<1 ms
From 3.3.3.1: bytes=32 seq=4 ttl=126 time=15 ms
From 3.3.3.1: bytes=32 seq=5 ttl=126 time=16 ms

--- 3.3.3.1 ping statistics ---
  5 packet(s) transmitted
  4 packet(s) received
  20.00% packet loss
  round-trip min/avg/max = 0/11/16 ms
```

查看会话表

❖ 在PC-a上使用Ping命令后，立即查看FW1上的会话表

```
[test1-a-policy-security]disp firewall session table verbose
2023-04-03 15:03:41.550
 Current Total Sessions : 4
 icmp  VPN: public --> public  ID: c387ff2504aeb48483642aeabb
 Zone: trust --> untrust  TTL: 00:00:20  Left: 00:00:19
 Recv Interface: GigabitEthernet1/0/1
 Interface: GigabitEthernet1/0/0  NextHop: 2.2.2.2  MAC: 00e0-fc0c-553e
 <--packets: 1 bytes: 60 --> packets: 1 bytes: 60
 1.1.1.1:19943 --> 3.3.3.2:2048 PolicyName: default

 icmp  VPN: public --> public  ID: c487ff2504aec2025d2642aeabc
 Zone: trust --> untrust  TTL: 00:00:20  Left: 00:00:20
 Recv Interface: GigabitEthernet1/0/1
 Interface: GigabitEthernet1/0/0  NextHop: 2.2.2.2  MAC: 00e0-fc0c-553e
 <--packets: 1 bytes: 60 --> packets: 1 bytes: 60
 1.1.1.1:20199 --> 3.3.3.2:2048 PolicyName: default
```

END